

Task 4. Lights Out

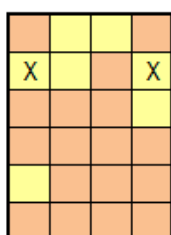
Available marks: 23

The classic *Lights Out*[®] game, marketed by Tiger Electronics in 1995, consists of a 5 by 5 grid of press-button lights. Each light is either on or off. When you press any light it changes state, from on to off, or off to on. In addition, each of its neighbours up, down, left and right changes state too. Lights on the edges have three neighbours, corner lights have only two.

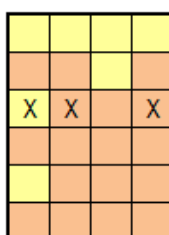
The version of the game used for this task has six rows and four columns. Given some starting combination of lights, the aim of the game is to determine which lights have to be pressed to turn them all off. The order in which the buttons are pressed has no effect on the final state. Fortunately algorithms exist to solve the puzzle, that is, to determine the buttons to press.

A fundamental algorithm (though not a complete solution) is called "chasing the lights". Starting with the second row, press each button directly *below* one that's currently on. Continue this process until the end of the last row, which may still contain some lights that are on.

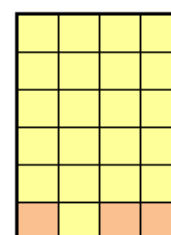
Example (yellow lights are off):



Press the two
lights marked
X, giving:



Press the lights
marked X, then after
processing the rest
of the rows you get:



How do you get rid of the pesky lights on the bottom row? It turns out that if you press a particular combination of lights on the *top* row, *before* chasing the lights, the last row will be all off and the puzzle completely solved. The correct top-row press pattern is dependent only on the bottom-row lights after they're chased down.

You are allowed to try all 16 combinations of button presses on the top row to find out which one solves the puzzle. The same pattern can solve all puzzles that produce the same set of bottom lights after the original chasing algorithm is applied. However, for full marks (see **Part B** overleaf) you will have to discover and report the corresponding top pattern for all 16 possible bottom light patterns.

In summary,

Chase the lights, remember the bottom row pattern for Part B
For each possible top-row press combination
 Restore the original configuration
 Apply the top row presses
 Chase the lights, remembering which lights were pressed
 If the bottom row lights are all off,
 Display the solution and exit

continued over...

Your Task

Write a program that reads a file containing a light pattern, using the notation where * is a light on and – is a light off. The first line contains the number of rows and columns, this is always 6 4.

The program must show any 6x4 pattern of light presses that fully solves the puzzle, using the notation **O** to represent a light being pressed and . to represent a light not pressed.

The input and correct output corresponding to the example above are:

6 4	
* _ *	00 . .
_ _ *	0 . 00
*** _	. 00 .
****	00 . .
_ ***	0 . 00
****	. . 00

Assessment

16 data files are provided, task4a.txt to taskp.txt. Each one produces a different bottom row on the first pass, so you can use them to explore the complete solution-generating range. You will be asked to show your solution for three randomly selected puzzles. Each of these is worth **6 marks** (total **18 marks**).

Partial marks (**5 only**) are possible if your program just chases the lights once and correctly displays the state of the bottom row.

Part B

Note that chasing the lights of the original configuration produced the bottom row light pattern "* _ **". Using a conventional binary notation, that's the decimal number 11 (8+2+1). As you can see from the example, the top pattern "00.." (12 if interpreted in binary) is incorporated into this solution, and into the solution of every puzzle that produces the same bottom row. This (and the 15 other combinations) could be used to limit the solution search to only two passes over the data, though your program is *not* required to do this. However you should use your program to help you discover these 16 solution-generating pairs.

The final **5 marks** are for identifying all solution-generating pairs and writing them in the table below (one is trivial and one was revealed above).

Bottom row light pattern (decimal):	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Top row solution (decimal):	0											12				